

EDA Notebook

0. Setup Environment

```
# DO NOT MODIFY THE CODE IN THIS CELL
```

```
!pip install -q utstd
```

```
from utstd.folders import *  
from utstd.ipyrenders import *
```

```
at = AtFolder(  
    course_code=36106,  
    assignment="AT3",  
)  
at.run()
```

```
import warnings  
warnings.simplefilter(action='ignore')
```

```
12.9/12.9 MB 38.6 MB/s eta
```

```
0:00:00
```

```
4.9/4.9 MB 74.0 MB/s eta
```

```
0:00:00
```

```
ERROR: pip's dependency resolver does not currently take into account  
all the packages that are installed. This behaviour is the source of  
the following dependency conflicts.
```

```
umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-  
learn 1.5.2 which is incompatible.
```

```
hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn  
1.5.2 which is incompatible.
```

```
Mounted at /content/gdrive
```

```
You can now save your data files in:
```

```
/content/gdrive/MyDrive/36106/assignment/AT3/data
```

Student Information

```
# <Student to fill this section>
```

```
group_name = "Group 20"
```

```
student_name = "Ratnadeep Patra"
```

```
student_id = "26294153"
```

```
# Do not modify this code
```

```
print_tile(size="h1", key='group_name', value=group_name)
```

```
<IPython.core.display.HTML object>

# Do not modify this code
print_tile(size="h1", key='student_name', value=student_name)

<IPython.core.display.HTML object>

# Do not modify this code
print_tile(size="h1", key='student_id', value=student_id)

<IPython.core.display.HTML object>
```

0. Python Packages

0.a Install Additional Packages

If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

```
# <Student to fill this section>
```

0.b Import Packages

```
# <Student to fill this section>
import pandas as pd
import altair as alt

import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import os
```

B. Data Understanding

```
# Do not modify this code
try:
    df = pd.read_csv(at.folder_path / "product_sub_category.csv")
except Exception as e:
    print(e)
```

B.1 Explore Dataset

```
pd.set_option('display.max_columns', None) # Forcing pandas to display
all columns
pd.set_option('display.max_rows', None) # Forcing pandas to display
all rows
```

```
df.describe(include='all').T
```

```
{
  "summary": {
    "name": "\ndf",
    "rows": 3,
    "fields": [
      {
        "column": "count",
        "properties": {
          "dtype": "date",
          "min": "37",
          "max": "37",
          "num_unique_values": 1,
          "samples": [
            "37"
          ],
          "semantic_type": "\"\"",
          "description": "\"\"",
          "unique": 4,
          "properties": {
            "dtype": "date",
            "min": 4,
            "max": 37,
            "num_unique_values": 2,
            "samples": [
              4
            ],
            "semantic_type": "\"\"",
            "description": "\"\"",
            "column": "top",
            "properties": {
              "dtype": "string",
              "num_unique_values": 3,
              "samples": [
                "5332b46f-2ec0-453b-8237-c78051d4c7bf"
              ],
              "semantic_type": "\"\"",
              "description": "\"\"",
              "column": "freq",
              "properties": {
                "dtype": "date",
                "min": "1",
                "max": "14",
                "num_unique_values": 2,
                "samples": [
                  "14"
                ],
                "semantic_type": "\"\"",
                "description": "\"\""
              }
            }
          }
        }
      }
    ]
  },
  "type": "dataframe"
}
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 37 entries, 0 to 36
```

```
Data columns (total 3 columns):
```

#	Column	Non-Null Count	Dtype
0	product_subcategory_id	37 non-null	object
1	product_category_id	37 non-null	object
2	name	37 non-null	object

```
dtypes: object(3)
```

```
memory usage: 1020.0+ bytes
```

```
df['name'].sample(10)
```

```
8      Derailleurs
29     Fenders
36  Tires and Tubes
32      Lights
20     Jerseys
15  Touring Frames
16      Wheels
1   Road Bikes
31  Hydration Packs
17    Bib-Shorts
```

```
Name: name, dtype: object
```

```
dataset_insights = ""
```

```
This dataset contains information about product subcategories, with 37
```

```
entries and 3 columns. There are no missing values across any of the
columns. The
columns are : 'product_subcategory_id' - unique identifier for each
product subcategory (acting as primark key); 'product_category_id' -
identifier for
the parent product category (4 unique values); and 'name' - the name
of the product subcategory (37 unique values). The dataset essentially
maps unique
product subcategories to their respective product categories.
"""
```

```
# Do not modify this code
print_tile(size="h3", key='dataset_insights', value=dataset_insights)

<IPython.core.display.HTML object>
```

B.2 Explore Feature of Interest product_category_id

```
for csv_file in os.listdir(at.folder_path):
    if csv_file.endswith('.csv'):
        try:
            if 'product_category_id' in pd.read_csv(at.folder_path /
csv_file).columns:
                print(csv_file)
            except UnicodeDecodeError:
                print(f"Warning: Could not read {csv_file} with default
encoding. Skipping or consider specifying encoding.")
            except Exception as e:
                print(f"Error reading {csv_file}: {e}")
```

```
product_category.csv
product_sub_category.csv
```

```
# <Student to fill this section>
feature_1_insights = """
The 'product_category_id' is a categorical feature with 4 unique
values. It serves as a foreign key to the 'product_category.csv'
dataset, where further
details about each product category can be found.
"""
```

```
# Do not modify this code
print_tile(size="h3", key='feature_1_insights',
value=feature_1_insights)

<IPython.core.display.HTML object>
```